Roll No.: .....

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech — Semester: IV

Name of the Course: Java Programming — Course Code: TCS408(E)

Time: 3 Hours — **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**

ii. Answer any two sub questions among a, b & c in each main questions

iii. Total marks in each main question are twenty.

iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the concept of 'Write Once, Run Anywhere' in Java. List and explain the different types of variables in Java.	CO1
(b)	A user is designing a banking system with a BankAccount as a base class and a SavingsAccount subclass. BankAccount has a default constructor and a constructor that takes accountNumber. SavingsAccount has a constructor that takes accountNumber and balance. Write a program in Java by implementing the concept of constructor chaining appropriately and show the memory initialization sequence when creating a SavingsAccount object.	CO1
(c)	(i) What will be the output of the given code. Give your answer with proper explanation.	CO2

A.

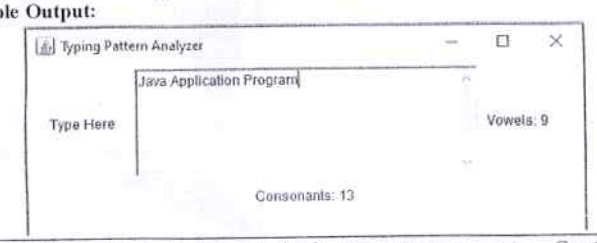
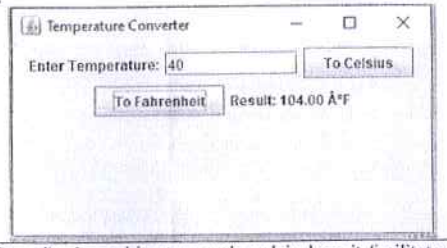
```
public class Test
 {
   static int x = 10;
   public static void main(String[] args)
   {
       Test t1 = new Test();
       t1.x = 20;
       Test t2 = new Test();
       t2.x = 30;
      System.out.println(t1.x);
       System.out.println(t2.x);
       System.out.println(Test.x);
   }
}
```

B.

```
interface A
 {
   static void print( )
   {
       System.out.println("A");
   }
}
class B implements A
{
}
public class Test
 {
   public static void main(String[] args) {
      B obj = new B();
      obj.print();
   }
}
```

(ii) List and explain the steps to create a custom immutable class in Java with example.

Q2	(20Marks)	
(a)	Differentiate between following with example: (i) Abstract class and interface (ii) throw and throws	CO2
(b)	A hospital maintains data for different types of employees such as **Doctors, Nurses,** and **Technicians**. All of them must be able to return their employee details. Create an interface named **HospitalEmployee** with a method getDetails(). Implement the interface in classes Doctor, Nurse, and Technician where each class should include fields relevant to its role (e.g., specialization for doctors). Write a Java program to create instances of each class and call getDetails() via HospitalEmployee reference.	CO2
(c)	An employee management system needs to organize employee details by departments. The system will have packages for employee information, department details, and payroll processing. Create a package employee that contains a class Employee with fields employeeId, name, and a method getEmployeeInfo().Create another package department that contains a class Department with fields departmentId, departmentName, and a method getDepartmentInfo(). Create one ore package payroll that contains a class Payroll with fields basic_salary, HRA, DA, gross_salary, and a method calculateSalary() to calculate total salary of employee. In the Main class, use the classes from all three packages to create objects and display the employee details. **Assume:** HRA=10% of basic_salary, DA= 15% of basic_salary	CO2
Q3	(20Marks)	
(a)	Compare and contrast the final, finally, and finalize() features in Java. Discuss their purpose, usage, and execution flow in the context of an example.	CO3
(b)	Write a Java program to accept bank account creation details: account number, holder name, and account type. Raise a **customized checked exception** in the following situations: • The holder name does not start with a capital letter. • The account number is not exactly 10 digits. • The account type is not one of: "Savings", "Current", "Salary"	CO3
(c)	A user wants to book a ticket, but the booking system is still loading available seats. The booking thread must wait until loading is done. Create a Java program using inter-thread communication where the ticket booking thread waits for the loading thread to fetch available seats. When seats are loaded by loading thread then ticket booking thread will book a ticket and prints " Seats were available. Booking confirmed! "	CO3
Q4	(20Marks)	
(a)	A store keeps track of its inventory in a **Product.txt** file which contains product id, product name, price and quantity. Write a program in Java which is able to perform following operations on **Product.txt** file: (i) Write operation to store information of 5 different products (ii) Read data from the file and display to the user.	CO3

(b)	Write a Java program using a Vector to manage a list of books in a library. Each book should have the following attributes: ID, Title, Author, and Availability. The program should perform the following operations: • Add new books to the library. • Remove a book by its ID. • Display all books in the library.	CO4
(c)	(i) Differentiate between AWT and Swing components in Java. (ii) A store requires a form to collect product details for generating a bill. Create an AWT GUI application (without event handling) structured as follows: • Top Panel: Label ***"Product Billing Form"***. • Center Panel: Form with Labels and TextFields for ***Product Name, Product ID, Quantity, Unit Price.*** • Bottom Panel: Buttons: ***Calculate, Reset, Print Bill.***	CO4, CO5
Q5	(20Marks)	
(a)	Create an application with a typing area and use KeyListener to count the number of vowels and consonants typed. Display the counts live below the input area. **Sample Output:** 	CO6
(b)	A weather application requires a simple temperature converter. Create a GUI application with: A JTextField to enter a temperature. Two buttons: "To Celsius" and "To Fahrenheit". A JLabel to display the result. Handle ActionEvents for both buttons and implement the given conversion logic. Temperature in degrees Fahrenheit (°F) = 9/5 (Temperature in degrees Celsius (°C) + 32) Temperature in degrees Celsius (°C) = 5/9 (Temperature in degrees Fahrenheit (°F) - 32). **Sample Output:** 	CO6
(c)	Define JDBC. Describe its architecture and explain how it facilitates communication between a Java application and a database. What package must be imported to use JDBC in Java?	CO6